Peng Ruoxin

M.Sc. Student · Artificial Intelligence and Big Data Computing

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RESEARCH PROFILE

My research focuses on efficient reuse and composition of pretrained vision models. I am interested in variable-cardinality model reassembly, low-cost architecture evaluation, multi-objective search, and computer-vision systems for scientific and agricultural applications.

EDUCATION

2025–2027 **The Hong Kong Polytechnic University**
M.Sc. in Artificial Intelligence and Big Data Computing · Hong Kong SAR, China

- Expected graduation: March 2027

2021–2025 **China Agricultural University**
B.Eng. in Data Science and Big Data Technology · Beijing, China

- GPA: 3.76/4.00 · Rank: 5/29

RESEARCH EXPERIENCE

2025–Present **HeteroWeave: Variable-Cardinality Vision Model Reassembly**
Research project · Three-person team

- Contributed to parallel block-composition rules, low-cost candidate ranking, and Pareto-guided search across accuracy, parameters, and FLOPs.
- Developed CLAS, a cardinality-aware activation proxy whose ranking correlations reach 0.775 and 0.570 on two validation pools.
- Evaluated the framework across image classification, vision-language retrieval, semantic segmentation, and object detection; the selected ImageNet-1K model reaches 79.457% Top-1 with 26.7% fewer parameters and 11.9% fewer FLOPs than serial single-block reassembly.

2022–2024 **Machine-Vision Rice Classification and Quality Evaluation**
Research project · China Agricultural University

- Built and annotated approximately 20,000 images across six rice varieties and developed detection, variety-classification, completeness-grading, and chalkiness-analysis modules.
- Produced a co-first-author paper accepted at the AAAI 2026 AgriAI Workshop and Rice Quality Detection Software V1.0.

2023–2024 **Computer-Vision Analysis of Pig Aggregation Behavior**
Research project · China Agricultural University

- Created approximately 3,000 annotated video frames and combined individual detection with spatial-neighborhood analysis to estimate aggregation.
- Published as first author at ECPLF 2024 and delivered Pig Aggregation Degree Calculation Software V1.0.

PUBLICATIONS

An Overall Real-Time Mechanism for Classification and Quality Evaluation of Rice
Wanke Xia*, **Ruoxin Peng***, Haoqi Chu, Xinlei Zhu, Zhiyu Yang, Yiting Zhao, Lili Yang
AAAI 2026 Workshop on AI to Accelerate Science and Engineering in Agriculture. Accepted. *Equal contribution.

An Improved Pure Fully Connected Neural Network for Rice Grain Classification
Wanke Xia, **Ruoxin Peng**, Haoqi Chu, Xinlei Zhu, Zhiyu Yang, Lili Yang, Bo Lv, Xunwen Xiang
International Conference on Computer Applications and Information Technology. Accepted at CAIT 2025.

Intelligent Recognition of Live Pig Aggregation Behavior Based on Convolutional Neural Network
Ruoxin Peng, Chao Liang, Peiguang Xin
European Conference on Precision Livestock Farming. First author, ECPLF 2024.